

Lab Assignment #06

Aim:

Design a microstrip line at a given resonant frequency and analyze the results.

Theory:

A microstrip line is a planar transmission line structure consisting of a conducting strip separated from a continuous ground plane by a dielectric substrate. It is widely used in microwave circuits due to its low profile, ease of fabrication, and compatibility with printed circuit technology.

The fundamental operation of a microstrip line is based on quasi-TEM wave propagation. Unlike rectangular waveguides, microstrip lines do not support a pure TEM mode because part of the electromagnetic field propagates in the dielectric substrate while another portion exists in the surrounding air. This results in an effective dielectric constant (ϵ_{eff}) that lies between unity and the relative permittivity (ϵ_r) of the substrate material.

For this experiment, Alumina (99.5%) was selected as the substrate material, having a relative permittivity $\epsilon_r \approx 9.9$. Due to the high dielectric constant, the guided wavelength (λ_g) within the substrate is significantly reduced compared to free space.

The characteristic impedance (Z_0) of the microstrip line depends primarily on the strip width (W), substrate thickness (h), and dielectric constant (ϵ_r). For proper impedance matching with standard microwave measurement systems, the line is typically designed for 50Ω . When the line is correctly matched, reflections are minimized and the reflection coefficient (S_{11}) approaches zero, resulting in a Voltage Standing Wave Ratio (VSWR) close to unity.

Resonance in a microstrip structure occurs when standing waves are formed along the conductor length.

Scattering parameters (S-parameters) are used to characterize the behavior of microwave circuits. The parameter S_{11} represents the reflection coefficient at the input port, while S_{21} represents the forward transmission coefficient. A minimum in S_{11} or VSWR indicates optimal impedance matching and minimal reflection at that frequency.

Farfield radiation patterns can also be obtained from the structure. Although a microstrip line is not intended to function as an antenna, radiation may occur due to edge effects, discontinuities, and finite ground plane dimensions. The radiation characteristics can therefore provide insight into leakage and unintended radiative behavior of the transmission line.

Table 1 - Parametric Details of Microstrip Line

Parameter	Value
Substrate Material	Alumina (99.5%)
Relative Permittivity (ϵ_r)	9.9
Substrate Thickness	0.5 mm

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Strip Width	0.479 mm
Conductor Thickness	0.01 mm
Board Size	30 mm
Frequency Range	1 - 6 GHz

Procedure:

1. Open CST Studio Suite and select the New Project option from the main dashboard.
2. Select the Microwave & RF / Optical template and choose the Planar Structures or General EM simulation environment.
3. Select the Time Domain Solver to analyze broadband S-parameters efficiently across the specified frequency range.
4. Set the project units by configuring Dimensions to mm and Frequency to GHz.
5. Define the Frequency Range with a Minimum of 1 GHz and a Maximum of 6 GHz to analyze the transmission characteristics within the required band.
6. Open the Modeling tab and configure the Background properties. Set the boundaries to Open (add space) in all directions to allow proper field propagation and radiation observation.
7. Open the Parameter List window and define the following parameters:
 - a. PEC_thick = 0.01 mm
 - b. strip_width = 0.479 mm
 - c. board_thick = 0.5 mm
 - d. epsilon_r = 9.9
 - e. board_size = 30 mm
8. Create the dielectric substrate using the Brick tool with dimensions defined by board_size and board_thick. Assign the material properties corresponding to Alumina ($\epsilon_r = 9.9$).
9. Create the ground plane at the bottom surface of the substrate using PEC material.
10. Create the microstrip conductor on the top surface of the substrate using the defined strip_width and PEC thickness.
11. Navigate to Macros → Calculators → Transmission Line Impedance and use the built-in macro to calculate the characteristic impedance of the microstrip line based on substrate thickness, dielectric constant, and strip width. Verify that the calculated impedance is approximately $50\ \Omega$ before proceeding with the simulation.
12. After confirming the impedance, define the excitation ports by navigating to Macros → Solver → Ports. Use this option to automatically generate the appropriate waveguide ports at both ends of the microstrip line, ensuring proper reference to the ground plane.
13. Navigate to the Simulation tab and define S-parameter monitors across the full frequency range.
14. Add a Farfield Monitor at the frequency corresponding to the minimum S11 value observed during simulation (5.97 GHz).
15. Open Setup Solver, verify all solver settings, and click Start to run the Time Domain simulation.

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16. Navigate to the 1D Results folder to extract S11, S21, and VSWR plots.
17. Navigate to the Farfields folder within the 2D/3D Results section to observe the radiation characteristics at the selected frequency.

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3D Model:

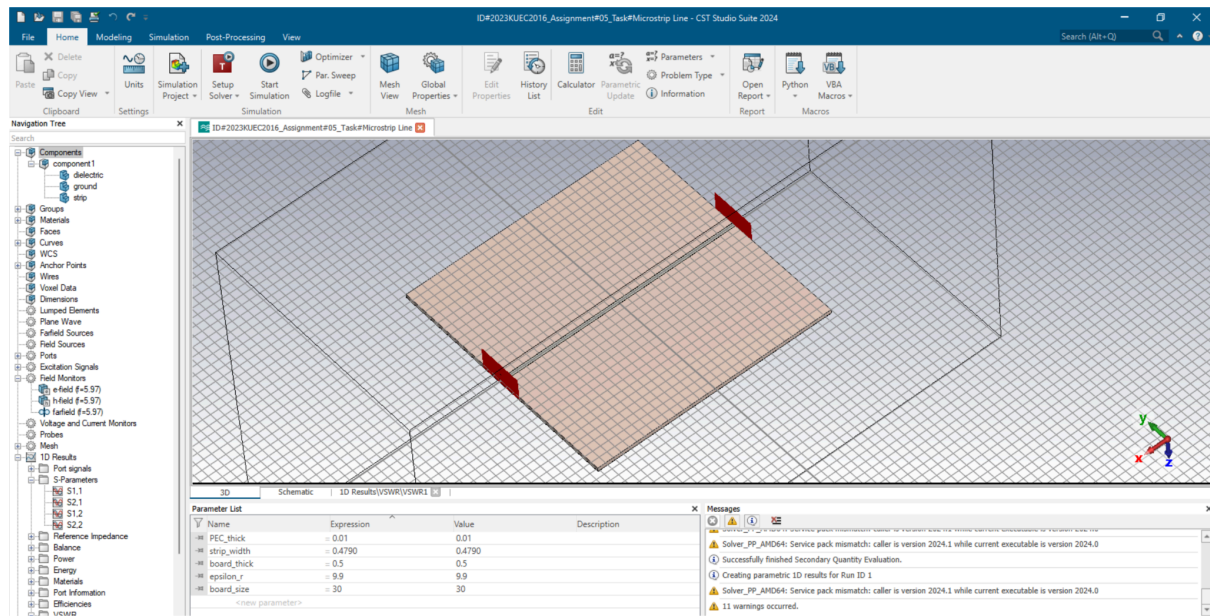


Fig 1.1 3D Model of Microstrip Line

Results:

Below are the results for simulating EM waves through a Microstrip Line over the range of frequencies 1 to 6 GHz.

Scattering Parameters

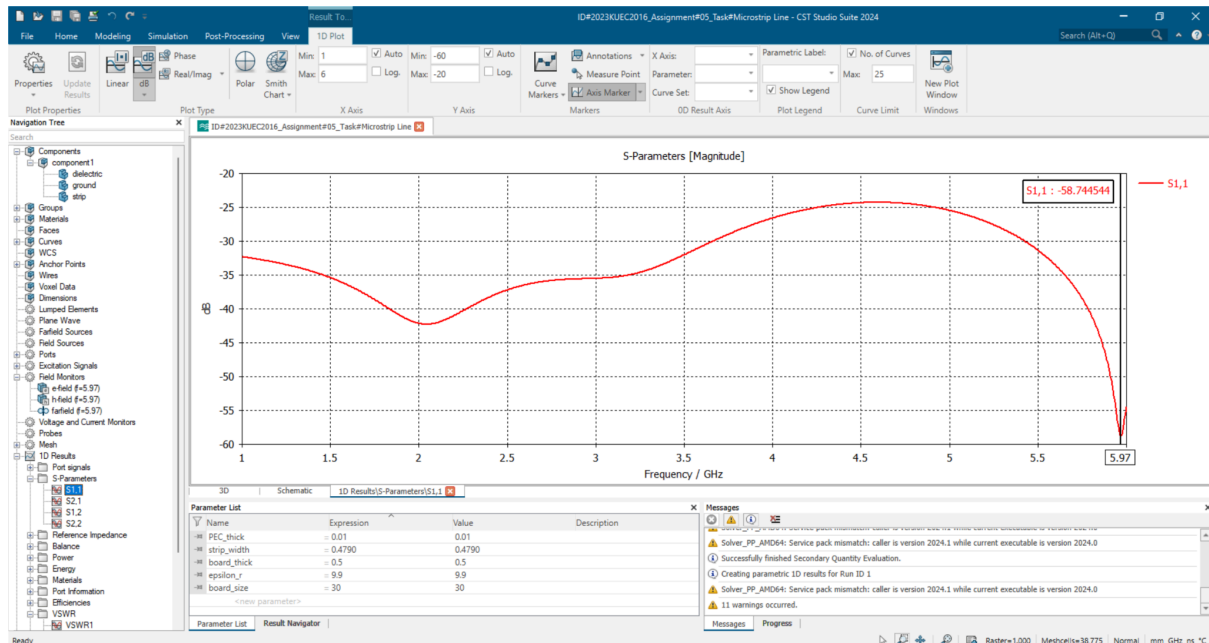


Fig 2.1 Scattering parameter $S(1,1)$, minimum value found at 5.97 GHz, which will be used to simulate resonant frequency to obtain maximum possible gain

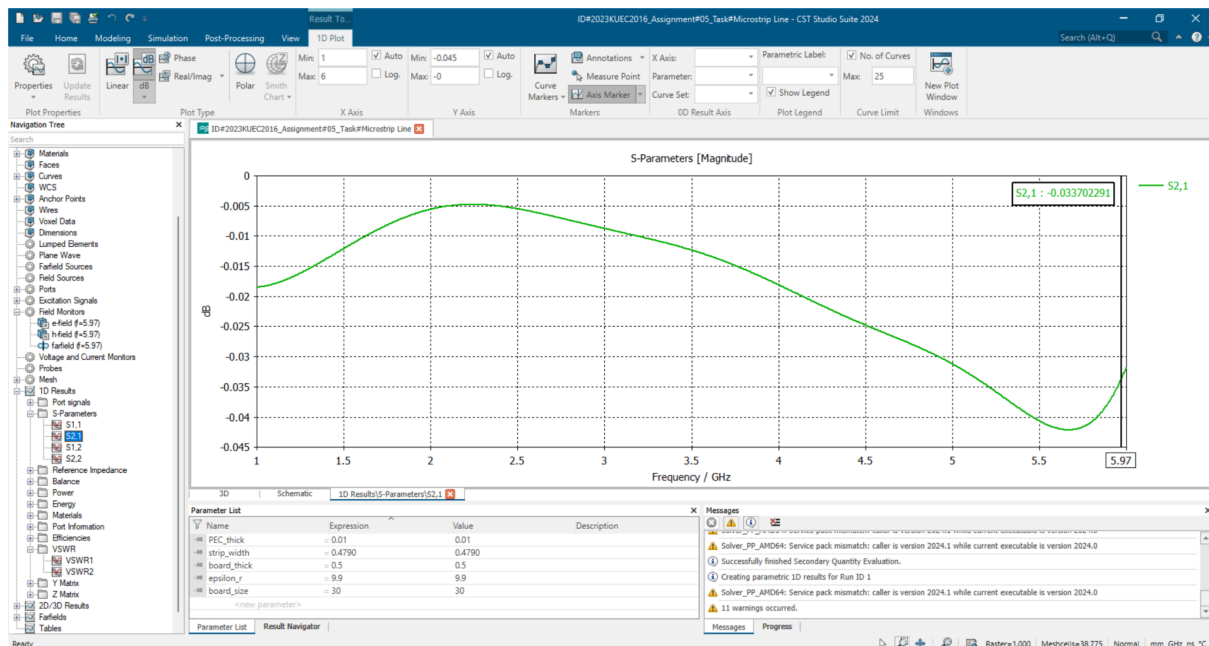


Fig 2.2 Scattering parameter $S(2,1)$

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VSWR

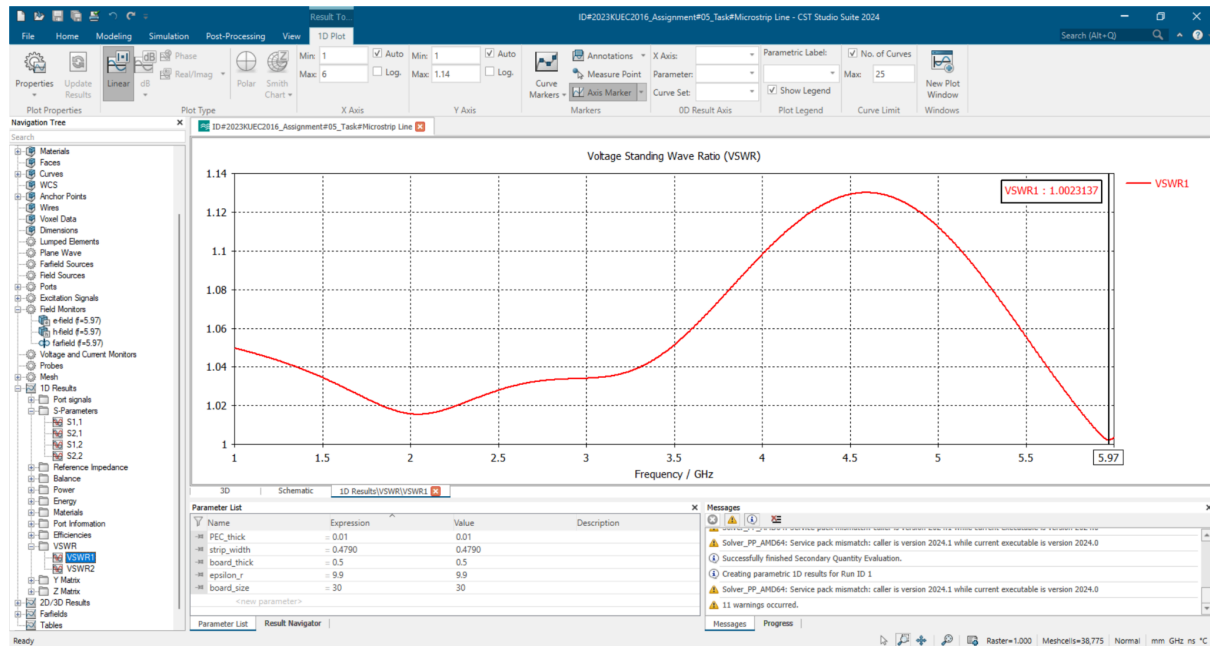


Fig 3.1 Voltage Standing Wave Ratio (VSWR) found minimum at $f = 5.97$ GHz

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Farfield Characteristics

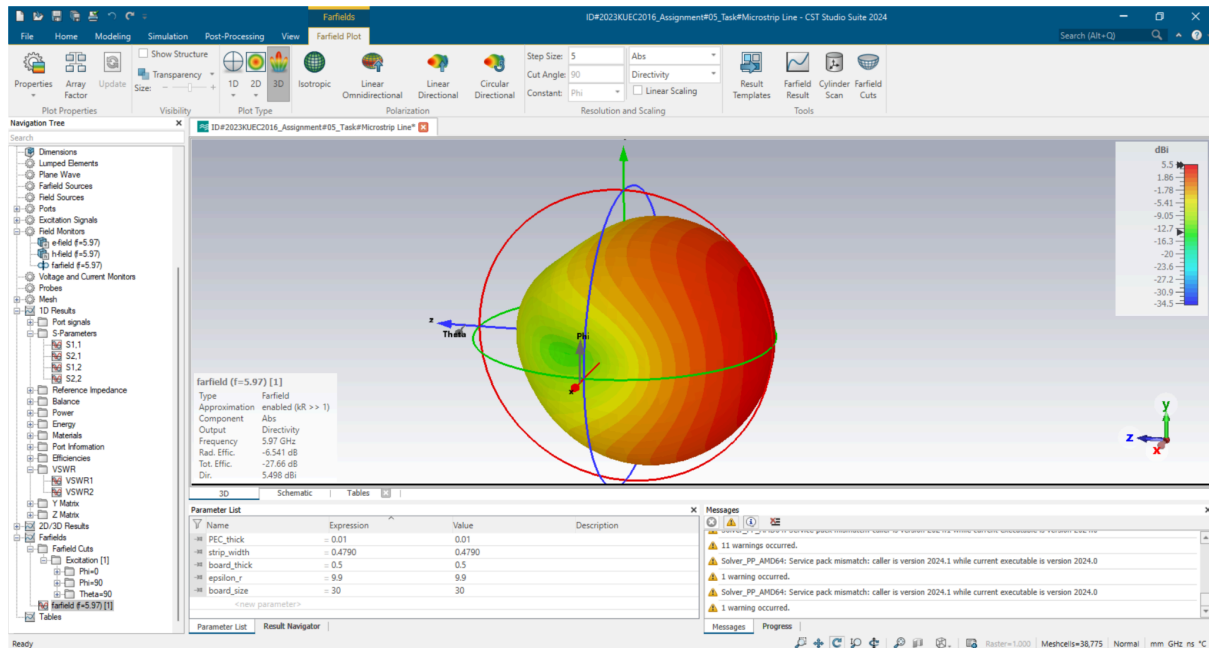


Fig 4.1 3D view of Farfield Distribution at resonant frequency $f = 5.97$ GHz

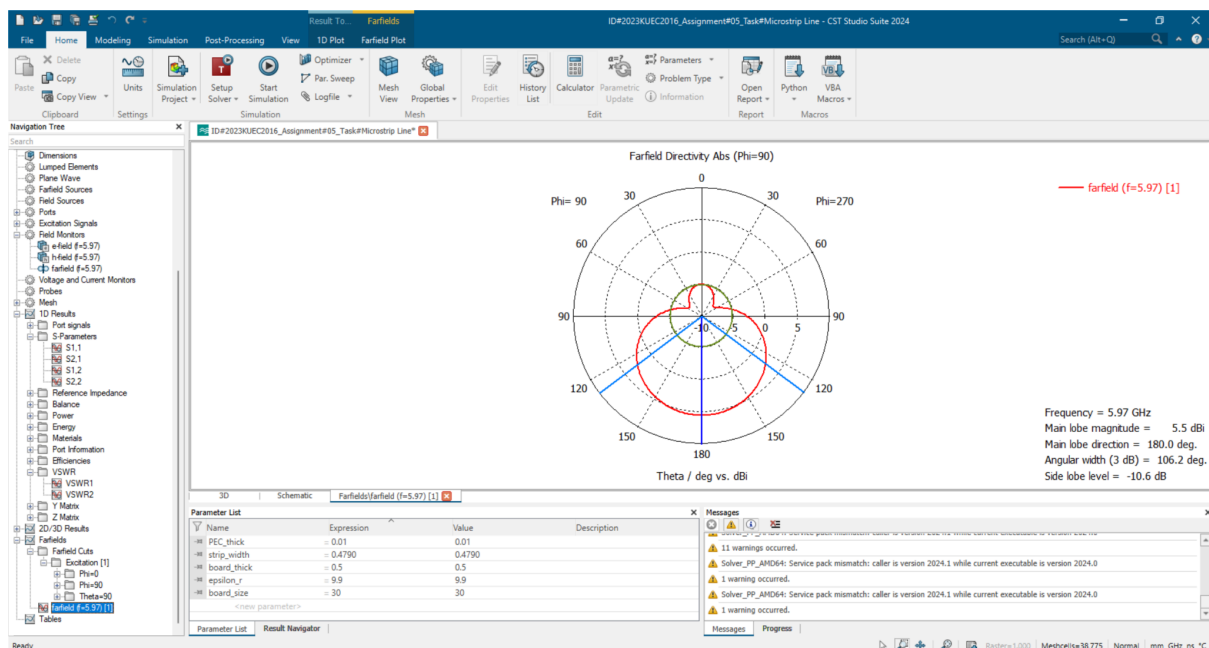


Fig 4.2 1D view of Farfield Distribution at resonant frequency $f = 5.97$ GHz

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Observations

The simulation of the microstrip line on Alumina (99.5%) across the frequency range of 1 to 6 GHz demonstrated excellent impedance matching behavior. Analysis of the S11 scattering parameter revealed a minimum reflection at 5.97 GHz. This frequency corresponds to the point of optimal impedance matching within the defined frequency band.

The VSWR plot further confirmed this observation, with the minimum VSWR occurring at 5.97 GHz and approaching unity. A VSWR value close to 1 indicates negligible reflection and efficient power transfer between ports. The overall VSWR variation across the band remained low, confirming broadband matching performance of the designed transmission line.

The forward transmission coefficient S21 showed stable transmission characteristics across the band, indicating minimal insertion loss and effective signal propagation through the microstrip structure.

Farfield analysis at 5.97 GHz revealed a broad radiation pattern with a main lobe magnitude of approximately 5.5 dBi directed around 180 degrees. Since the structure is primarily a transmission line rather than a radiating antenna, the observed radiation is attributed to edge diffraction, finite ground plane effects, and substrate surface-wave leakage. The beamwidth of approximately 106 degrees confirms a wide radiation pattern typical of unintended microstrip radiation.

Overall, the structure behaves predominantly as a matched transmission line rather than a resonant radiating element.

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Conclusions:

The design and simulation of the microstrip transmission line on an Alumina (99.5%) substrate successfully demonstrated efficient guided-wave propagation within the 1 to 6 GHz frequency range.

The structure achieved optimal impedance matching at 5.97 GHz, as confirmed by the minimum S11 reflection coefficient and near-unity VSWR. The results indicate that the microstrip dimensions were appropriately selected to approximate a 50 Ω characteristic impedance.

Transmission performance remained stable across the frequency band, with minimal insertion loss observed in the S21 parameter. Farfield radiation analysis showed moderate leakage radiation, which is expected due to the finite dimensions of the substrate and conductor.

Unlike resonant antennas or cavity structures, the microstrip line primarily functions as a guided transmission medium. The results confirm that the structure efficiently transfers electromagnetic energy between ports with minimal reflection, validating the design methodology and simulation setup.

The experiment successfully illustrates the principles of microstrip transmission line design, impedance matching, and S-parameter analysis using CST Studio Suite.